

Harish Kumar S

AI/ML Engineer

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<https://github.com/Harish-nika>

<https://harish-nika.github.io/>

PROFESSIONAL SUMMARY

Integrated MTech in Data Science (VIT, 2025) AI/ML engineer building production systems at FactEntry Data Solutions (A SIX Company). Shipped the Data Mining Engine on K3s with GitLab CI, Argo CD GitOps, Redis queue workers, and FastAPI microservices. Also delivered RAG document systems, PyPI-published NLP tooling, Roboflow computer-vision models, and containerised AI deployments.

- Production K3s bond-extraction platform (80+ CPU cores, Redis streams, human-in-the-loop validation)
- GitOps: GitLab CI -> Helm promote -> Argo CD sync to stage
- RAG & document intelligence: FAISS, Groq LLaMA, Streamlit/FastAPI
- PyPI package: language-processing-tool | Roboflow YOLOv11 dental CV models

CORE COMPETENCIES

AI/ML Engineer | Machine Learning | Deep Learning | Computer Vision | NLP | RAG | Python | FastAPI | Kubernetes | K3s | Docker | Redis | PostgreSQL | GitLab CI | Argo CD | Helm | GitOps | MLOps | Roboflow | YOLO | FAISS | LangChain | Streamlit

EXPERIENCE

Software Developer Trainee

FactEntry Data Solutions - A SIX Company | Dec 2025 - Present

- Shipped Data Mining Engine (DME) modules on K3s stage: corporate/municipal bond extraction, validation UI, and cluster operations portal.
- Implemented Redis-stream job dispatch with multi-node worker throughput and GitOps delivery (GitLab CI image build -> Helm chart promote -> Argo CD).
- Built FastAPI microservices, live WebSocket job telemetry, role-based CPU tiering, and domain-restricted enterprise auth.

MLOps and Data Science Intern

FactEntry Data Solutions - A SIX Company | Nov 2024 - May 2025

- Published language-processing-tool on PyPI (PDF language detection, OCR, document clustering).
- Built Fixed Income Expert RAG system (FAISS, Groq LLaMA 3, Streamlit/FastAPI) for bond document Q&A.
- Developed MedBot medical PDF RAG prototype with SBERT embeddings and grounded LLM responses.

Associate Software Engineer Trainee Intern

MIMASOFT Technologies | Jun 2025 - Oct 2025

- Contributed to backend integration, API workflows, and production-oriented software development practices.

EDUCATION

Integrated MTech - CSE (Data Science), Vellore Institute of Technology (2020 - 2025) - CGPA 7.8

Class XII, THIRU G V C Higher Secondary School (2020) - 79%

Class X, St Joseph Matriculation Higher Secondary School (2018) - 87.8%

TECHNICAL SKILLS

AI/ML: Machine Learning, Deep Learning, Generative AI, Computer Vision, NLP, RAG, FAISS, Sentence Transformers, Ollama, Roboflow, YOLOv11, Supervision

Engineering: Python, FastAPI, Streamlit, SQL, Docker, Kubernetes (K3s), Redis, PostgreSQL, LangChain, PyMuPDF, OpenCV

DevOps/GitOps: GitHub, GitLab CI, Argo CD, Helm, systemd, GitHub Actions, GitOps

PROJECTS

Data Mining Engine (DME) - K3s, FastAPI, Redis, PostgreSQL, Argo CD

Enterprise bond extraction on K3s - UI, API, Redis queue workers, GitOps CI/CD, validation workspace.

<https://harish-nika.github.io/dme-project.html>

Fixed Income Expert System - FastAPI, Streamlit, FAISS, Groq, SBERT

RAG chatbot for financial bond PDFs - upload, embed, retrieve, LLaMA 3 answers via Groq.

<https://harish-nika.github.io/fexpert.html>

Cybersecurity Content Moderator - LLM, Streamlit, Docker, FastAPI

Capstone: multimodal hate-speech and harmful content classification (text, image, PDF).

<https://harish-nika.github.io/content-moderator.html>

Language Distribution & Document Clustering Tool - Python, Tesseract, LangDetect, PyPI

PyPI-published PDF language detector, OCR, and clustering pipeline (v0.2.7).

<https://harish-nika.github.io/lang-tool.html>

MedBot (Medical RAG Assistant) - Streamlit, FAISS, Sentence Transformers, Groq

Medical PDF Q&A with SBERT + FAISS retrieval and Groq LLaMA 3 grounded responses.

<https://harish-nika.github.io/medbot.html>

<https://github.com/Harish-nika/medbot>

Dental ML Project - Roboflow, YOLOv11, inference-sdk, Supervision

YOLOv11 tooth & panoramic X-ray detection - Roboflow fine-tuning, serverless and local inference CLI.

<https://harish-nika.github.io/dental-project.html>

https://github.com/Harish-nika/Dental_ML_project

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals
- Applied Data Science (SmartInternz + Google Developers)
- Data Analysis and Visualization with Power BI
- Preparing Data for Analysis with Microsoft Excel
- Harnessing the Power of Data with Power BI
- Extract, Transform, and Load Data in Power BI